

Did the Computer Hear It Right?

The answer key, how we built it, and what we test next

PSYCH-ASR

Part 1 — Three models and a piece of arithmetic

Two robots work on the recording, and a third step glues their answers together.

Who	What they do
the typist	writes down every word that was said
the stopwatch	marks exactly when each word starts and stops
the name-tagger	decides which of the two people was talking, and when

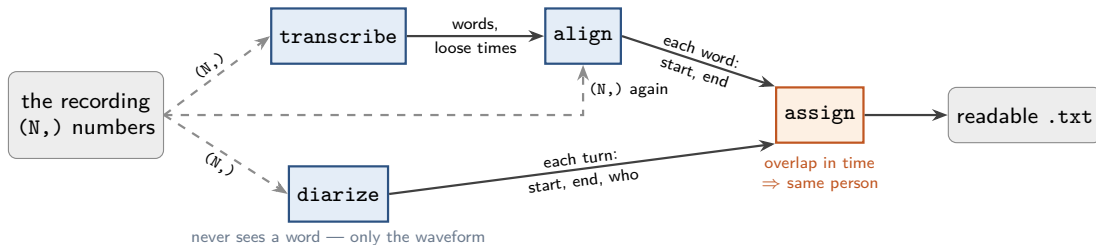
Put them together and you get **subtitles with names on them**:

THERAPIST: So how was your week?

PATIENT: It was okay I guess.

One line there is one “turn” — a single unbroken stretch of one person talking, ending when the other person starts. That box holds **two turns**. The whole deck is about getting the turns right, so it is the one word worth keeping.

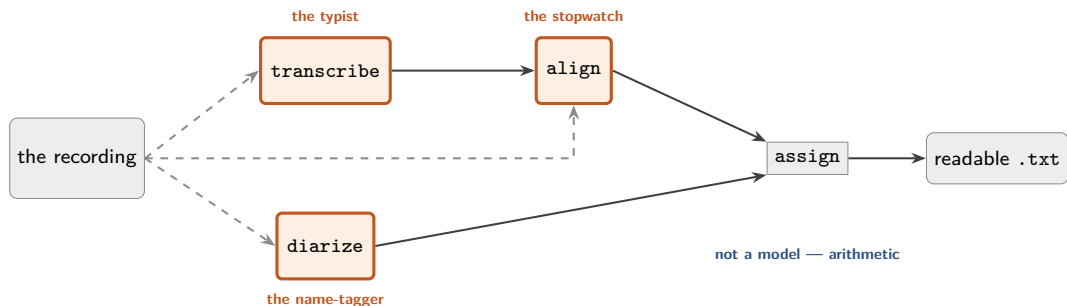
How the pieces fit together



Three boxes do real work: transcribe is **the typist**, align is **the stopwatch**, diarize is **the name-tagger**. assign is not a model — it is arithmetic on the two lists of times.

Three of those boxes are choices

`assign` is arithmetic and has no alternatives. The other three are **models we picked**, and every one of them can be swapped for a different model.



Wrong words, wrong times, or wrong names. Three separate failures, three separate models to test.

To say one model beats another you need to know what the *right* answer was. That is what Madison built.

She opened the machine's transcript as a plain text file with **numbered lines**, put the audio on, and listened to the whole session. Every time the machine got something wrong, she added **one row** to a spreadsheet.

She is not writing a transcript from scratch. She is recording the *difference* between what the machine said and what she heard.

117 rows. Fifty minutes of listening.
That spreadsheet is the entire ground truth this project has.

What is in one row

Column	What it holds
Timestamp	where she heard it
Line	the line number in the machine's text file — her only pointer into the machine's own version
Speaker	who <i>really</i> said it: Therapist or Participant
Error	one of six labels (next slide)
AI Transcript	what the machine wrote there — or the word <code>None</code> if it wrote nothing
Actual Speech	what was really said — or <code>None</code> if nothing was
Severity	her judgement of how much it matters
Add Turn?	TRUE if fixing this creates a turn the machine never produced

Two columns carry the weight. **Line** says *where*; **Add Turn?** says whether the *words* or the *conversation's shape* went wrong.

The six kinds of error

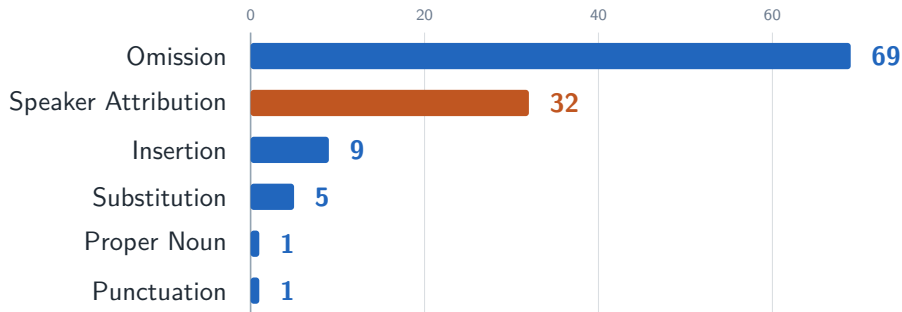
Label	What went wrong
Omission	something was said and the machine wrote nothing
Insertion	the machine wrote something nobody said
Substitution	the machine wrote the wrong words
Proper Noun	a name or place came out mangled
Punctuation	the words are right, the sentence breaks are not
Speaker Attribution	the words are right and they are filed under the wrong person

The first five are **the typist** making mistakes. The last one is **the name-tagger** making mistakes — the machine heard perfectly and pointed at the wrong person.

32 of the 117 rows are Speaker Attribution. Those are the rows that grade the part of the pipeline the whole project depends on.

How often each kind happened

Six labels, 117 rows — and two of the labels are almost the whole sheet.



■ the typist mis-hearing ■ the name-tagger mis-filing

69 omissions and **32** misattributions are **101** of the 117 rows. Punctuation and mangled names happen **once each**.

Three ways the machine can be wrong

Every one of the 117 rows is one of these three shapes. Invented exchanges; T is the therapist, P the participant.

	What the machine produced	What was really said
Turn added	T did it help	T did it help P not really
Turn moved	T did it help not really	T did it help P not really
Words only	P it was tuesday	P it was thursday

The first two want the same answer. “Not really” is the participant’s own turn either way. They differ only in what the machine did with it: in the first it never wrote those words at all, in the second it wrote them inside the therapist’s turn. Both fixes end with **one more turn** on the page than the machine produced.

Only the third leaves the turns alone. The first two change **who spoke, and when** — and that is what every structural feature we plan to measure is counted from.

Most of the sheet is about turns, not spelling

One column, **Add Turn?**, is ticked when fixing the row would *add or move a turn* — the first two shapes — rather than just change the words inside one. Split the 117 rows on that tick:

What the machine did	Rows
Missed something a person said, completely	47
Put one person's words inside the other's turn	27
Wrote down something nobody said	1
Turns wrong — ticked	75
Wrote the wrong words; the turns were right	42
Every row in the sheet	117

75 of the 117 rows are *turn added* or *turn moved*. Only the **42** are *words only*.

Every one of the 117 rows becomes exactly one edit. The next six slides are the whole algorithm: **her row, the machine's turns before, the turns after.**

There are only four things the program can do, and **Add Turn?** is what picks between them:

Operation	What changes	Used
replace	the words inside one turn	35
insert	a new turn appears	53
extract	words move out into a turn of their own	21
relabel	the name on a turn	5

Only **replace** leaves the turns alone. The other three are why this is not find-and-replace.

Example 1 — the machine misheard a word

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
132	Participant	Substitution	really tired lately	really wired lately	FALSE

BEFORE

PARTICIPANT [12:04-12:09] i've been really tired lately

AFTER

PARTICIPANT [12:04-12:09] i've been really wired lately

replace. One turn in, one turn out. Same speaker, same times, three characters different. **Proper Noun** and **Punctuation** rows do exactly this too — a mangled name, a sentence broken in the wrong place.

Example 2 — words dropped out of the middle of a turn

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
88	Therapist	Omission	<i>None</i>	for the whole week	FALSE

None leaves no words to search for, so Line 88 is the only anchor — and it names a *line* of the file, not a turn. Her own numbered view:

BEFORE

```
87 [08:15] THERAPIST
88 so you did that every day
      ^ the end of line 88's words
```

AFTER

```
87 [08:15] THERAPIST
88 so you did that every day for the whole week
```

replace at a single point: **the end of line 88's words**. Nothing in the sheet says where inside the line they go — that is assumed, and it is the one thing this row cannot pin down. **Add Turn?** is **FALSE**, so the shape of the conversation is unchanged.

Example 3 — something somebody said, missed completely

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
205	Participant	Omission	<i>None</i>	mm hmm	TRUE

BEFORE

THERAPIST [19:40-19:58] and that's the part we want to build on next week

AFTER

THERAPIST [19:40-19:51] and that's the part we want to build on

PARTICIPANT [19:51-19:51] mm hmm

THERAPIST [19:51-19:58] next week

insert. Now **Add Turn?** is **TRUE**, so a turn that never existed appears — and the therapist's turn is cut in two around it. **One turn became three.** There were **47** rows like this.

Example 4 — two people welded into one turn

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
48	Participant	Speaker Attribution	yeah Therapist	yeah Participant	TRUE

BEFORE

THERAPIST [04:12-04:31] did you try the walk yeah so how did that go

AFTER

THERAPIST [04:12-04:20] did you try the walk

PARTICIPANT [04:20-04:22] yeah

THERAPIST [04:22-04:31] so how did that go

extract. The words were transcribed perfectly — they were filed under the wrong person. **Look at the two text cells: both end in a role name.** That is her convention, and it is how the program knows the machine said “Therapist” where the truth was “Participant”.

Example 5 — the right words under the wrong name

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
301	Therapist	Speaker Attribution	how did that feel Participant	how did that feel Therapist	FALSE

BEFORE

PARTICIPANT [28:03-28:07] and how did that feel

AFTER

THERAPIST [28:03-28:07] and how did that feel

relabel. Same words, same times, one name changed. **Add Turn? is FALSE** because the machine did find a turn here — it just put the wrong person's name on the front of it. Only **5** rows are this simple.

Example 6 — the machine wrote something nobody said

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
156	Participant	Insertion	thank you for watching	<i>None</i>	FALSE

BEFORE

PARTICIPANT [14:22-14:30] i guess so thank you for watching

AFTER

PARTICIPANT [14:22-14:30] i guess so

replace, with nothing. This one is worth recognising: “thank you for watching” is a phrase **the typist** learnt from video subtitles and emits into **silence**. It is not a mishearing — there was no sound there at all.

The case that needs care — two rows, one “yeah”

Madison logs a misattribution **twice**, from both sides:

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
48	Participant	Speaker Attribution	yeah Therapist	yeah Participant	TRUE
Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
49	Therapist	Insertion	yeah	<i>None</i>	FALSE

Row 48 says *move that “yeah” to the participant*. Row 49 says *take that “yeah” out of the therapist’s turn*. **They describe the same three characters** — and moving them *is* taking them out.

Run both and the word is deleted twice. So the first edit **claims** those characters, and row 49 is reported as *already handled* — not applied, and not a failure either. One row on this sheet ended up in that pile.

The result

	Machine	Corrected
Turns	89	215
Words	7,298	7,388
Speaker labels	anonymous	Therapist / Participant

Of the 117 rows: **114** applied, 1 already handled by an earlier row, and 2 the program refused to place — it could not find her words, so it **changed nothing** and named the rows.

The machine heard **89** back-and-forths. There were **215**.
It loses the short ones — and the short ones are most of the conversation's shape.

The sheet also names the speakers, for free

Look at Example 4's row again. The **AI Transcript** cell says what the machine decided:

Line	Speaker	Error	AI Transcript	Actual Speech	Add Turn?
48	Participant	Speaker Attribution	yeah Therapist	yeah Participant	TRUE

The machine's own label for that turn is `SPEAKER_00`. Madison wrote that the machine called it *Therapist*. So `SPEAKER_00` **is** the therapist — stated, not guessed. Across the sheet: **49** rows say so, 3 disagree.

And it **inverts** the obvious guess. In the corrected transcript the therapist holds **78.6%** of the talk time — session 1 is the didactic intro. “The therapist talks less” would have got this session exactly backwards.

What this reference cannot do — look at Example 3 again

AFTER

THERAPIST	[19:40-19:51]	and that's the part we want to build on
PARTICIPANT	[19:51-19:51]	mm hmm
THERAPIST	[19:51-19:58]	next week

That “mm hmm” lasts **zero seconds**, and 19:51 is a guess — the split fell where the *characters* divide, not the sound.

So it can grade

Were the words right?

Was the speaker right?

Did the machine find the turn at all?

It cannot grade

Were the times right?

Where exactly a turn starts and stops

Anything measured in seconds

Where inside a turn a word belongs

Fixing the *times* is one job: run **the stopwatch** over the audio using Madison's corrected words as the script. Then every word has a measured time.

...and it cannot place words inside a line

Line names one *wrapped* line of the file, not a whole turn. A long turn is several numbered lines:

```
[12:30] PARTICIPANT
41    i went for a walk and felt better
42    so i tried it again on sunday
```

Invented lines, narrowed to fit the slide; the real file wraps at 96 characters.

Madison logs Line 41, **Omission**, *None*, “with my dog”. Those words were really said just after “walk” — in the *middle* of line 41. But *None* leaves nothing to search for, so the only anchor is the line:

Where they belong	i went for a walk with my dog and felt better
Where they land	i went for a walk and felt better with my dog

Same turn, same speaker, same words — only the *position* is wrong. That costs the diarizer score nothing and the reference text a little. **58** of the 117 rows are placed by line number alone.

Part 4 — Three models, and only one has been compared

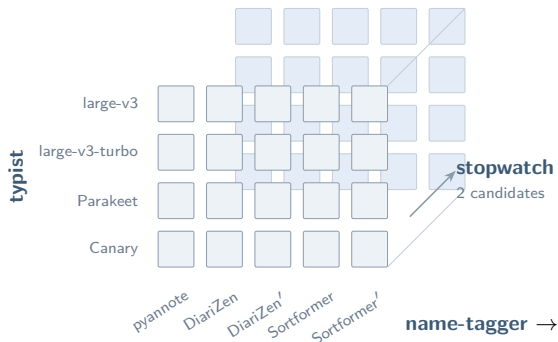
Box	Nickname	What runs today	Compared?
transcribe	the typist	Whisper large-v3	no
align	the stopwatch	wav2vec2 base, trained on read audiobooks	no
diarize	the name-tagger	five candidates	yes — undecided

The stopwatch is the uncomfortable one: it is the *small* model, trained on 960 hours of people **reading books aloud**. Therapy speech is spontaneous, overlapped and disfluent.

Now that there is an answer key, all three can finally be graded.

The grid has three axes, not two

Three boxes are swappable, so the sweep is a **cube**: 4 typists × 2 stopwatches × 5 name-taggers = 40 cells.



Still cheaper than it looks. the name-tagger never reads a word, so its answer depends on neither of the other two. 4×2 runs that write the words and time them, plus 5 that only listen — **13 jobs, not 40** — and the cells are assembled from the pieces.

Each axis can only move some of the numbers

The words come from the typist. The times come from the stopwatch, timing the typist's words. Who-spoke comes from the name-tagger, which never reads a word.

Swap this	Words change?	Times change?	So it can move
the typist	yes	yes	every kind of error
the stopwatch	no	yes	who-spoke only — the times are what hand each word to a speaker
the name-tagger	no	no	who-spoke only

Fix the typist and **five of the six error counts are already decided** for that whole row of the grid — one measurement per row plus one per cell, not six per cell. The *counts*, not their turn-or-words split: whether a missing utterance was a whole turn depends on where the name-tagger put its boundaries.

Part 5 — 39 more cells, and nobody is going to listen to them

Madison's pass produced **one** cell of that cube. It cost fifty minutes of listening and 117 typed rows.

What she did, once	What we need, 39 more times
Listened to the recording	nobody has fifty minutes $\times$ 39
Compared it to the machine's words	Compare it to <i>her corrected transcript</i> instead
Wrote down each error and its kind	Count the same kinds, the same way

The corrected transcript is the recording's stand-in. Every question she answered by listening — was this word right, was this the right person, did the machine even find this turn — can be answered by comparing two pieces of text.

Grade against the answer key, not against the audio.
Fifty minutes becomes two seconds, and it runs on a laptop.

The whole algorithm, in three steps

Step	Why it is in this order
1 Line the two word lists up, ignoring who spoke	Which words match is the one question that must not be influenced by the speaker labels, or the grade starts assuming its own answer
2 Read off which cluster is whom , from the words that matched	SPEAKER_00 is nobody until its words agree with the key's. A word both sides wrote is the only place the two naming schemes can be compared
3 Ask two questions of every word	<i>was it written down, and as itself</i> — then, of the words that were, <i>is it under the right person</i>

Step 3 is why this is not just a word error rate. A single percentage cannot tell an arm that mistyped forty words apart from an arm that filed forty *correct* words under the wrong person — and those break different things downstream.

No timestamp is ever read. The key's turn times are interpolated guesses (slide 21), so a grade that used them would be grading the guess. **Words and names only**, which is exactly the part of the key that is trustworthy.

One session, one arm, four differences

Invented lines. T is the therapist, P the participant; the arm's own speaker names are still numbers, because that is all a name-tagger ever produces.

THE ANSWER KEY

T [00:00-00:06] so how was your week and did the activity schedule help at all
P [00:06-00:06] mm hmm
P [00:06-00:12] it was fine i went for a walk with my dog on thursday
T [00:12-00:16] tell me more about Madison and what she said

WHAT THIS ARM PRODUCED

SPEAKER_00 so how was your week and did the activity schedule help at all it was fine
SPEAKER_01 i went for a walk with my dog on tuesday
SPEAKER_00 tell me more about Madeline and what she said

Four things are wrong, and only two of them are spelling. The arm lost mm hmm entirely, and it filed it was fine — which it transcribed *perfectly* — inside the therapist's turn.

What the grader called them

Label	Key says	Arm said	Turns wrong?	Found by
Omission	mm hmm	--	yes	key words with nothing opposite
Speaker Attribution	it was fine	it was fine	yes	words matched, the <i>name</i> did not
Substitution	thursday	tuesday	no	one word standing for another
Proper Noun	Madison	Madeline	no	same, but capitalized <i>mid-sentence</i>

5 of the 6 labels fall out of lining the words up. The sixth — *Speaker Attribution* — is found among the words that **matched**, which is why a grade that stopped at the word error rate would call this transcript good.

The only judgement: did the turns change?

Her **Add Turn?** column is not in the text — it is a question about the text. One rule answers it, and it is the same rule in both directions:

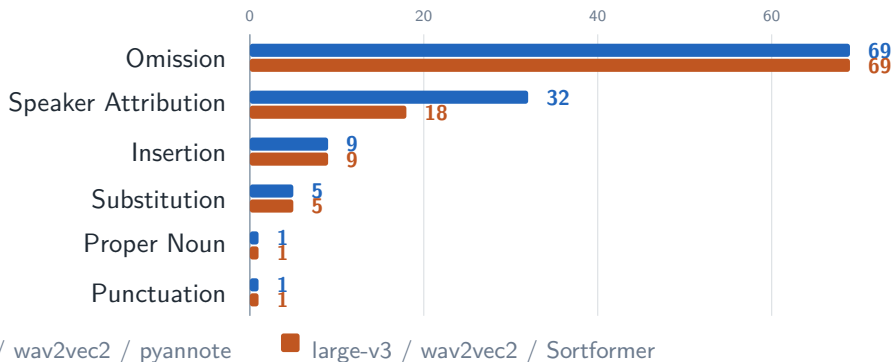
Is the *whole* turn involved, or only part of one?

	Every word of the turn	Only some of its words
Words missing	the machine never heard this turn — a turn appears	it heard the turn and mistyped it — words only
Wrong speaker	the boundary is right, the name is wrong — words only	fixing it cuts the machine's turn open
Words invented	a turn disappears	words only

In the example: mm hmm is the participant's whole turn, so a turn appears. it was fine is only part of the machine's first turn, so that turn gets cut open. **2** of the **4** findings change the turn structure — her ratio, computed.

What comes out: slide 8's chart, once per cell

Same typist and stopwatch, two different name-taggers. **Invented numbers** — no cell has been measured.



Five bars cannot move and one can. The typist is the same in both, so the same words are right and wrong in both. Swapping **the name-tagger** changes exactly one bar — and that is the bar two thirds of Madison's sheet was about.

What the computed grade cannot say

She recorded	Why a program cannot
Severity, Meaning Changed	A judgement about the session, not a property of the text. Two identical word swaps can matter enormously and not at all
One row per error	She logged one row per thing she decided had happened. This logs one finding per contiguous difference, so adjacent slips may split or merge
Anything in seconds	The key's times are interpolated (slide 21). Timing needs a measured reference, which is the next job on the list

So the computed profile is the *same kind* of number as hers, not the identical number.

The check that *is* free: grade the arm she actually read. The key is that arm plus her 117 corrections, so her sheet and the computed profile describe the same errors — and a classifier that disagrees with her about what an omission *is* shows up there, before any other cell is believed.

- 1 A person listened for fifty minutes and logged **117** errors. **75** of them are about *who spoke*, not about spelling.
- 2 Each row becomes one of four edits. **89** turns became **215**.
- 3 That transcript is the first thing here that can tell a model it is wrong — and it already inverted the guess we would have used to name the speakers.
- 4 Three models to test, not one. The grid is a cube: **40** cells from **13** jobs.
- 5 And nobody listens again. Every cell is graded against her transcript in her own categories, which is **39** more listening passes we do not have to ask for.

We built nothing on top of this transcript until somebody checked it.
Now that it is checked, it is what checks everything else.